

Aditya Kumar Singh

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SUMMARY

Senior Software Engineer with expertise in **C/C++ system programming**, **Java backend development**, and cloud-native distributed systems. Strong experience in **Linux internals**, **multithreading**, **memory management**, **concurrency**, **IPC**, and low-latency networking for 5G-CRAN and CPaaS platforms. Proficient in **Kubernetes**, **Docker**, and building scalable telecom and cloud systems handling high-concurrency workloads.

EDUCATION

- **Indian Institute of Technology (IIT) Kharagpur** May 2026 - Present
Executive Program in Applied AI & Machine Learning Kharagpur, India (Online)
- **Indian Institute of Technology (IIT) Hyderabad** 2019 - 2022
M.Tech in Computer Science Engineering | CGPA: 8.64 Hyderabad, India
- **University Institute of Engineering & Technology, CSJMU** 2014 - 2018
B.Tech in Computer Science Engineering | CGPA: 7.89 Kanpur, U.P.

EXPERIENCE

- **Radisys India Limited** Mar 2023 - Present
Senior Software Engineer - Systems & Cloud Infrastructure Bangalore
 - CPaaS & SIP Call Control (Java & Cloud)**
 - **RESTful SIP APIs:** Engineered Java-based REST APIs and integrated the **Cipango SIP stack** via Servlets for real-time call signaling (INVITE, PRACK, BYE), ensuring carrier-grade telecom workflows.
 - **SDP Parsing & Negotiation:** Developed **Java utilities** to extract and validate media IP/ports via SDP, resolving SIP interoperability and Via header compliance issues.
 - **Code Modernization & DevOps:** Refactored legacy Java codebases for multi-tenant environments and automated HTTP/HTTPS API deployments using **Shell scripts**.
 - 5G-CRAN Platform (C/C++ & Cloud)**
 - **DU Feature Development:** Designed and developed DU application features including Multi-FH and Multi-MH connections, improving network scalability and flexibility.
 - **Distributed Systems & Kubernetes:** Built a custom **Kubernetes controller in Go** for distributed 5G-RAN systems, reducing deployment effort by **30%** compared to Helm while enforcing CPU and HugePages allocation.
 - **System Performance Tuning:** Performed **Linux system-level optimization** for CU-DU modules by tuning CPU governors and I/O paths; diagnosed **NUMA bottlenecks** using **pidstat/numastat**, reducing CPU usage and power consumption by **15%**.
 - **Advanced Debugging & Stability:** Performed root-cause analysis using **GDB and AddressSanitizer (ASan)**, resolving memory corruption and concurrency-related issues.
 - **CI/CD & Code Quality:** Established a C++ unit testing framework achieving **90%+ code coverage**, preventing regression bugs in CI/CD pipelines.
- **Dremio Software India Private Limited** Aug 2022 - Feb 2023
Software Engineer - Cloud Platform Hyderabad
 - **Billing System Framework:** Designed and developed billing infrastructure for Dremio Cloud using **Java** and **MySQL** for high-volume data metering.
 - **Marketplace Integration:** Integrated billing accounts with **AWS Marketplace** via **TACKLE** for automated payment workflows.
 - **Query Acceleration:** Optimized BI/SQL query performance using **Data Reflections**, reducing latency by **30%**.
 - **Financial Integrity:** Engineered reconciliation logic to detect inconsistent replicas and automate credit reversals.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Golang, Java, Python, MySQL, Bash Scripting
- **Systems & Networking:** Linux Internals, Multithreading, Concurrency, IPC, Memory Management, Low Latency, Distributed Systems
- **Cloud & DevOps:** Docker, Kubernetes, Git, Maven, CMake, Helm, AWS Marketplace
- **Domain & Performance:** 5G-RAN, CPaaS, SIP, SDP, MRF, Intel VTune, Linux Perf, GDB, Wireshark, ASan

PROJECTS

•Master's Thesis & Intel Sponsored Research on O-RAN Optimization

Aug 2019 - Jul 2022

IIT Hyderabad

- **O-RAN Optimization:** Performed CAPEX/OPEX analysis for multiple CU-DU-RU split deployment scenarios in 5G O-RAN networks.
- **Open Source Contribution:** Contributed to **OpenAirInterface (OAI)** based 5G Cloud-RAN optimization for improved energy efficiency on Intel Xeon platforms, accelerating L1/L2 workloads using **Intel AVX-2** and **AVX-512** SIMD instructions.
- **Kernel Profiling:** Performed kernel and system profiling using **Intel VTune**, **Linux perf**, and **GDB** to optimize high CPU and power consumption paths.
- **Automation:** Optimized real-time OAI testbed performance using **USRP SDRs** and developed **Python** automation utilities for large-scale log analysis.

RESEARCH AND PUBLICATIONS

- **Conference Paper:** U. Pawar, **Aditya Kumar Singh**, et al., "Understanding Energy Consumption of Cloud Radio Access Networks: an Experimental Study," *2020 IEEE 3rd 5G World Forum (5GWF)*, 2020. [[Link](#)]
- **Poster Presentation:** Shah Alam, **Aditya Kumar Singh**, "Design and Implementation of Image Cryptography using RSA Algorithm," *IEEE International Conference on Engineering Science & Advance Research*, 2018.

CERTIFICATIONS

- **Architecting with Google Kubernetes Engine: Foundations** - Coursera
- **Data Reflections** - Dremio Software
- **Dremio Fundamentals** - Dremio Software

MISCELLANEOUS

- Competitive Programming:** Active problem solver on LeetCode, HackerRank, and GeeksforGeeks, focusing on Data Structures & Algorithms (DSA) and System Design concepts.
- Interests:** Cricket, Chess, Fitness Training